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BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: R26-DS-010

Project Title: Multimodal Computational Intelligence for Improving Caregiver Performance and Care Delivery

IT Number: IT22294098

Individual Component Title: Context-Aware Intelligent System for Automated ADL Extraction and Real-Time Documentation from Caregiver Voice Logs

	Identification and Analysis
Data Privacy Concerns	The component handles health-related information about older adults: medication, meals, hygiene, mobility, symptoms and vital signs. Sri Lanka's Personal Data Protection Act, No. 9 of 2022, treats health data as a special category of personal data. Handwritten care records from the Love & Elder Care Home were collected only with the permission of the care home's management. Care recipients' names and ages were removed when the information was typed, and each care recipient was given a code (P01 to P41). All care recipient data used in this component was therefore de-identified. Only this de-identified information was given to the large language model used to generate the ADL dataset. No names and no photographs of records were shared with any AI tool. No audio was recorded at the care home and no care recipient was interviewed. The test recordings were spoken by the author and volunteers. In the photographs included in the thesis, care recipients' names on the records were obscured and their faces were blurred, so that no individual can be identified. An ethics application was submitted to SLIIT, and only de-identified information was used while the decision was pending. Informed consent and ethics approval are required before any identifiable care recipient data or real caregiver voice recordings are used.
Data Security	Care recipients are identified in the system only by codes, not by names.

	Structured ADL records and alerts are stored in a Supabase (PostgreSQL) database, and audio clips are stored in Cloudflare R2 object storage, both within the shared CareSense data layer. The three trained models (Whisper-small with LoRA, BART-base and T5-small) run in the project's own FastAPI backend, so voice recordings are not sent to an external AI model provider for processing. The text-to-speech step currently uses the gTTS library, which relies on an external Google web service. A commercial deployment would need a licensed text-to-speech service instead. A real deployment must restrict access to authorised staff, store records and audio securely, and define clear data retention periods.
Bias and Fairness	The speech model was adapted on the Svarah corpus of Indian-accented English, not on Sri Lankan caregivers, so accuracy may differ for local accents and individual speakers. The Svarah split was made by utterance and not by speaker, so the reported word error rate (15.57%) may be optimistic for new speakers. The ADL dataset was generated with LLM support from one care home's information, so its sentences are more regular than real caregiver speech. Alert recall fell from 100% on the test split to 82.0% on a separate 200-case safety set with different phrasings. Drug names such as Losartan can be misrecognised by speech recognition, which could affect medication records. The system currently supports English only, so caregivers who speak Sinhala, Tamil or code-mixed speech are not yet supported. To reduce unfair outcomes, model selection prioritised safety-relevant cases, and field-based alert rules were added to reduce missed alerts (recall raised to 93.3% with no false alarms on the safety set).
Transparency and Accountability	Written reports and spoken handover summaries are produced from fixed templates using only the extracted fields, so the system does not generate unsupported clinical content. Alert decisions follow clear, documented rules based on the alert definition given by the caregivers. All methods, datasets, model settings and results are documented. Fixed random seeds (42) were used, and results were saved as CSV files so they can be checked and reproduced.

	Limitations are reported openly, including the gap between test-split and safety-set results and the fact that the improved rule-based figures may be optimistic. The use of generative AI tools was disclosed, and the author checked all generated records against the handwritten records and caregivers' information. The system is a documentation and alerting aid only. It does not diagnose, and nurses and supervisors remain responsible for care decisions. Emergencies must still be reported directly by staff.
Impact on Stakeholders	**Caregivers:** can record observations by voice instead of writing by hand, saving time and reducing missed observations. Errors in transcription or extraction could create incorrect records, so outputs should be reviewed. **Care recipients:** may benefit from more complete records, earlier attention to symptoms and clearer handover between shifts. A missed alert could delay care, which is why alert recall was treated as the most important measure. **Supervisors :** receive structured records and alerts on the dashboard. False alarms or missed alerts could affect their workload and decisions, so alerts support their judgement and do not replace it. **Care home management:** gains consistent and searchable digital records, but must ensure consent, secure storage and compliance with data protection requirements. **Families:** may benefit from safer and more continuous care for their relatives.